

Stoichiometry Questions

- a) Propane can undergo the following combustion reaction to form carbon dioxide and water: $\text{C}_3\text{H}_{8(g)} + 5 \text{O}_{2(g)} \rightarrow 3 \text{CO}_{2(g)} + 4 \text{H}_2\text{O}_{(g)}$. If three molecules of propane is combusted, how many molecules of carbon dioxide are formed?

b) How many molecules of oxygen gas were needed to react with the 3 molecules of propane?

c) If 16 molecules of water were formed during the reaction, how many molecules of propane were reacted?

d) How many molecules of oxygen are needed to react with 3.02×10^{21} molecules of propane?
- a) Iron can be produced by the reaction of iron (III) oxide with carbon monoxide as follows: $\text{Fe}_2\text{O}_{3(s)} + 3 \text{CO}_{(g)} \rightarrow 2 \text{Fe}_{(s)} + 3 \text{CO}_{2(g)}$. How many moles of iron are produced from 3 moles of iron(III) oxide?

b) If 16 moles of carbon dioxide gas are formed, how many moles of carbon monoxide were used?

c) If you wanted to make 7 moles of iron, how many moles of iron(III) oxide would be required and how many moles of carbon monoxide would be required?
- a) Ammonia gas is synthesized from nitrogen gas and hydrogen gas according to: $\text{H}_{2(g)} + \text{N}_{2(g)} \rightarrow \text{NH}_{3(g)}$. How many moles of ammonia are formed from 6 moles of hydrogen gas?

b) How many moles of hydrogen gas will react with 5.8 moles of nitrogen gas?

c) If you wanted to make 13.5 moles of ammonia how many moles of nitrogen is required?
- a) Zinc metal reacts with sulphur to form zinc sulphide as follows: $8 \text{Zn}_{(s)} + \text{S}_{8(s)} \rightarrow 8 \text{ZnS}_{(s)}$. How many moles of zinc sulphide are formed from 0.5 moles of sulphur?

b) What is the mass of the zinc sulphide formed in part a)?
- a) Sodium chloride is synthesized from sodium metal and chlorine gas according to: $\text{Na}_{(s)} + \text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$. If 10 g of sodium is reacted with chlorine gas, how many grams of sodium chloride are formed?

Answers:

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|------------------------------------|-----------------------------|---------------|---------------|
| 1. a) 9 molecules | 2. a) 6 moles | 3. a) 4 moles | 4. a) 4 moles |
| b) 15 molecules | b) 16 moles | b) 17.4 moles | b) 390.0 g |
| c) 4 molecules | c) 3.5 moles and 10.5 moles | c) 6.75 moles | |
| d) 1.51×10^{22} molecules | | | |
5. 25.43 g